

GROSSE ILE, MI, USA

WMO: 725373

Lat: 42.099N

Lon: 83.161W

Elev: 587

StdP: 14.39

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 54819

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.4	9.4	-5.5	4.2	6.2	0.3	5.7	11.6	24.9	23.5	22.7	23.6	7.9	240	0.483

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.4	88.9	74.3	85.7	73.4	82.4	71.8	77.7	84.7	76.0	82.6	74.6	80.8	9.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.2	135.1	81.7	73.4	126.8	79.9	72.5	123.1	79.1	41.5	84.5	39.8	82.5	38.6	81.2	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.8	18.6	16.7	DB	-0.8	93.7	6.7	3.6	-5.7	96.3	-9.6	98.4	-13.4	100.4	-18.3	103.1	(4)
(5)			WB	-1.7	80.4	6.3	2.1	-6.2	81.9	-9.9	83.1	-13.4	84.2	-18.0	85.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.1	26.7	28.6	36.8	48.6	60.1	69.6	74.1	72.8	66.1	53.8	42.1	32.2		
	DBStd	18.62	10.30	9.54	9.68	8.04	7.61	6.56	5.43	4.88	6.63	7.80	8.23	8.93		
	HDD50	2734	724	598	425	116	8	0	0	0	48	260	555			
	HDD65	5929	1189	1018	876	493	191	29	2	4	65	357	687	1018		
	CDD50	3128	0	0	15	76	322	587	748	705	484	166	23	2		
	CDD65	849	0	0	0	3	40	167	285	245	99	10	0	0		
	CDH74	5718	0	0	3	34	257	1102	2245	1574	473	30	0	0		
	CDH80	1428	0	0	0	4	53	291	649	346	82	3	0	0		
	WSAvg	7.6	9.0	8.4	8.4	8.7	7.1	6.5	6.1	5.8	6.3	7.6	8.3	8.5		
	PrecAvg	33.6	2.3	2.1	2.3	3.1	3.6	3.3	3.3	3.1	3.1	2.7	2.4	2.3		
(15) Precipitation	PrecMax	46.4	4.0	5.0	4.8	5.4	7.8	5.7	6.0	6.0	6.1	6.9	5.7	4.5		
	PrecMin	25.9	0.8	0.7	0.7	0.7	1.3	1.3	1.3	0.5	0.8	0.4	0.6	0.8		
	PrecStd	4.6	0.9	1.0	1.1	1.1	1.6	1.3	1.2	1.4	1.7	1.3	1.1	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.9	56.7	69.8	79.3	84.4	90.8	93.3	90.6	86.4	78.6	64.6	57.1		
		MCWB	51.7	50.6	60.5	62.3	67.6	72.9	74.6	75.3	74.0	68.1	56.6	53.7		
	2%	DB	46.4	47.9	62.6	71.7	81.1	86.3	90.0	87.1	81.9	72.7	60.9	52.0		
		MCWB	43.9	44.3	54.9	59.1	67.6	72.2	75.0	74.4	71.2	65.1	54.5	49.7		
	5%	DB	43.3	43.7	54.8	64.4	75.5	82.9	86.2	83.6	79.3	69.8	56.7	46.3		
		MCWB	40.6	39.8	48.9	55.0	65.1	71.3	74.2	73.5	69.7	63.2	51.7	43.7		
	10%	DB	39.1	39.4	50.0	61.1	72.4	80.9	83.2	81.3	76.1	65.7	54.1	43.5		
		MCWB	36.9	36.7	45.3	52.7	63.4	69.9	73.1	71.7	68.5	60.2	50.0	40.2		
	0.4%	WB	51.9	50.8	61.6	65.2	73.4	77.3	81.1	80.0	77.1	70.7	59.1	54.6		
		MCDB	53.1	54.6	68.2	74.7	80.8	85.6	88.0	87.4	83.5	76.0	62.6	56.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	45.5	45.4	55.3	61.1	70.2	75.4	78.3	77.1	74.2	67.4	55.4	49.2		
		MCDB	46.3	47.8	59.9	67.7	77.4	82.4	85.1	83.6	79.6	71.2	59.9	51.6		
	5%	WB	41.1	40.4	50.4	57.7	67.5	73.9	76.5	75.4	72.3	63.8	53.0	44.8		
		MCDB	43.7	43.4	54.6	64.4	74.0	80.7	83.3	81.2	77.1	67.2	55.9	46.3		
	10%	WB	37.6	37.5	45.7	53.9	65.2	72.0	74.8	74.0	70.3	61.2	50.1	41.0		
		MCDB	39.3	40.0	48.8	59.1	70.5	78.1	81.4	79.3	75.0	65.0	53.5	43.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.2	12.2	14.3	16.3	16.4	15.7	15.4	14.5	14.8	14.1	12.0	9.8		
		MCDBR	14.0	16.7	21.4	22.7	20.3	17.8	17.2	16.2	16.4	17.2	15.8	13.4		
		MCWBR	12.6	14.0	15.4	14.2	11.5	9.1	9.0	8.8	9.6	11.2	12.5	11.9		
		MCDBR	13.8	16.3	20.6	21.2	18.1	16.0	15.3	14.0	14.1	14.9	15.1	13.2		
	5% WB	MCWBR	13.0	14.5	16.1	15.0	11.7	9.2	9.2	8.6	9.3	10.7	13.0	12.7		
(40) Clear-Sky Solar Irradiance	taub		0.304	0.337	0.356	0.402	0.436	0.451	0.442	0.435	0.391	0.362	0.321	0.316		
	taud		2.404	2.292	2.339	2.241	2.191	2.197	2.247	2.256	2.351	2.411	2.492	2.414		
	Ebn at Noon		266	272	282	275	267	263	264	262	266	261	259	250		
	Edh at Noon		25	33	36	42	45	45	43	41	35	29	23	23		
(44) All-Sky Solar Radiation	RadAvg		472	751	1108	1465	1715	1926	1931	1680	1359	887	555	387		
	RadStd		48	93	107	125	147	124	100	65	116	74	74	34		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	-1.19	N/A	N/A	N/A	N/A	N/A	+335	+342	N/A	N/A
(47)	Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+160	N/A

Nomenclature: See separate page